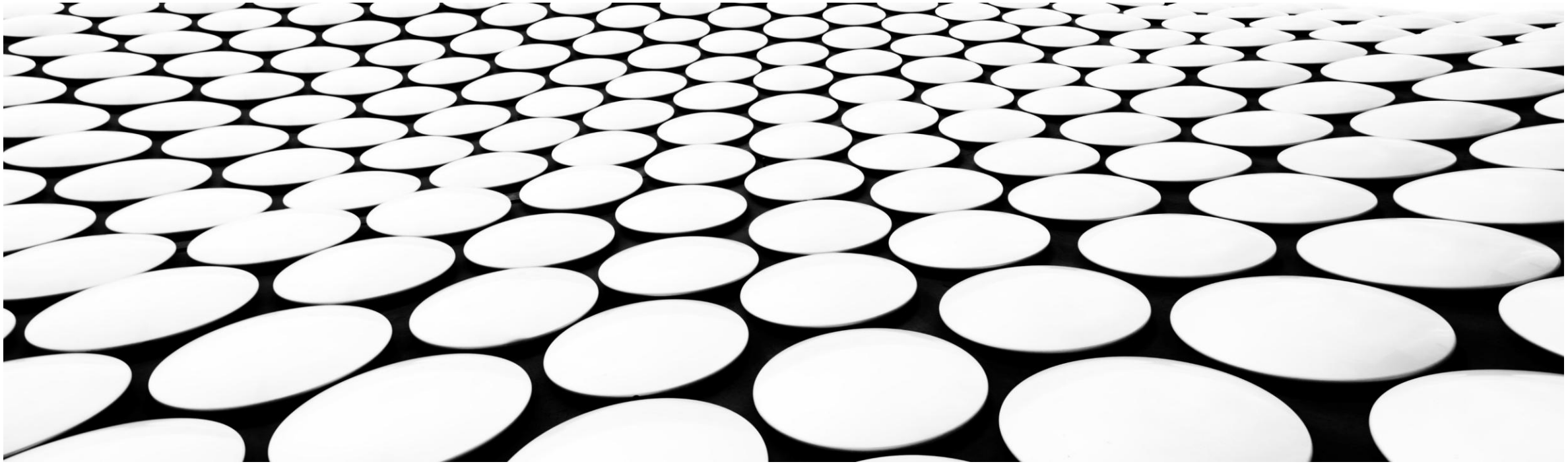
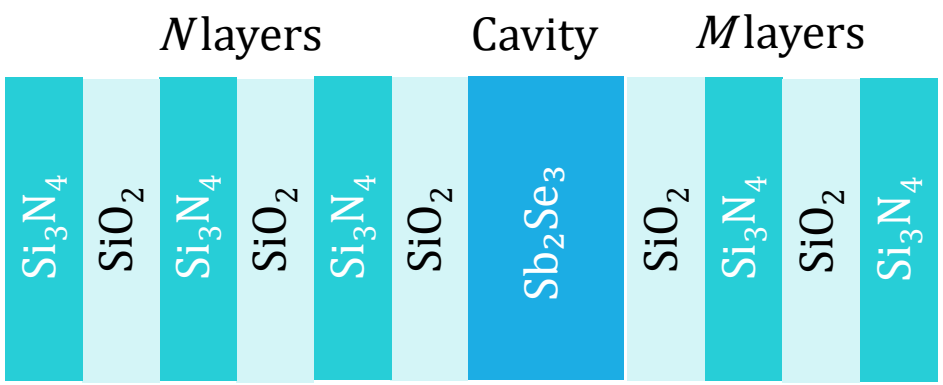
A Theoretical Study of Optical Limiter Designs Using Asymmetric Resonant Cavities with Sb_2Se_3 Layers — *Transfer-Matrix Method Approach* —

ADVANCED PHOTONIC LABORATORY

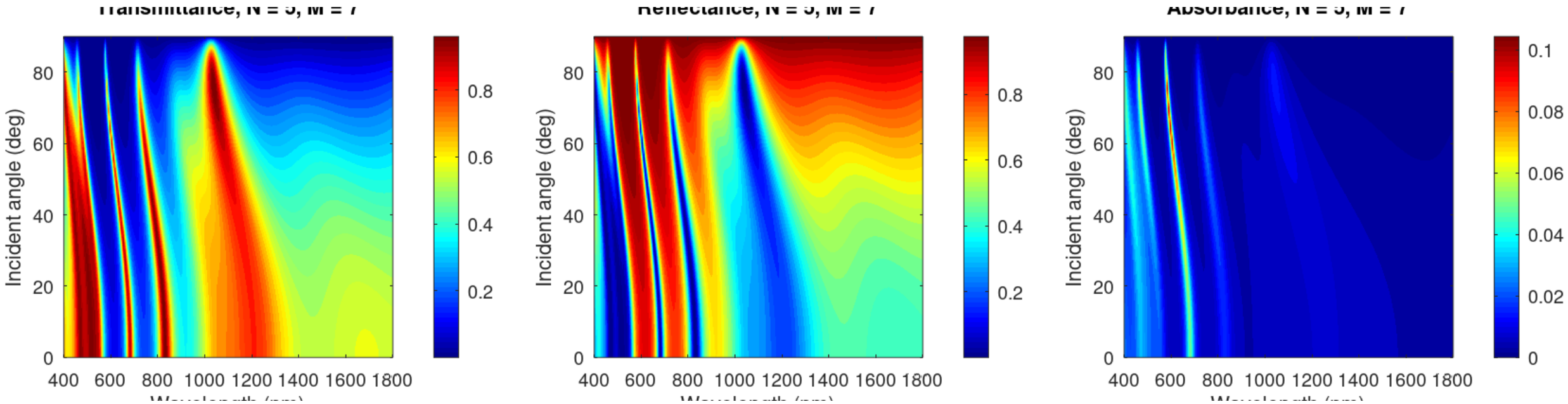


PRELIMINARY RESULTS

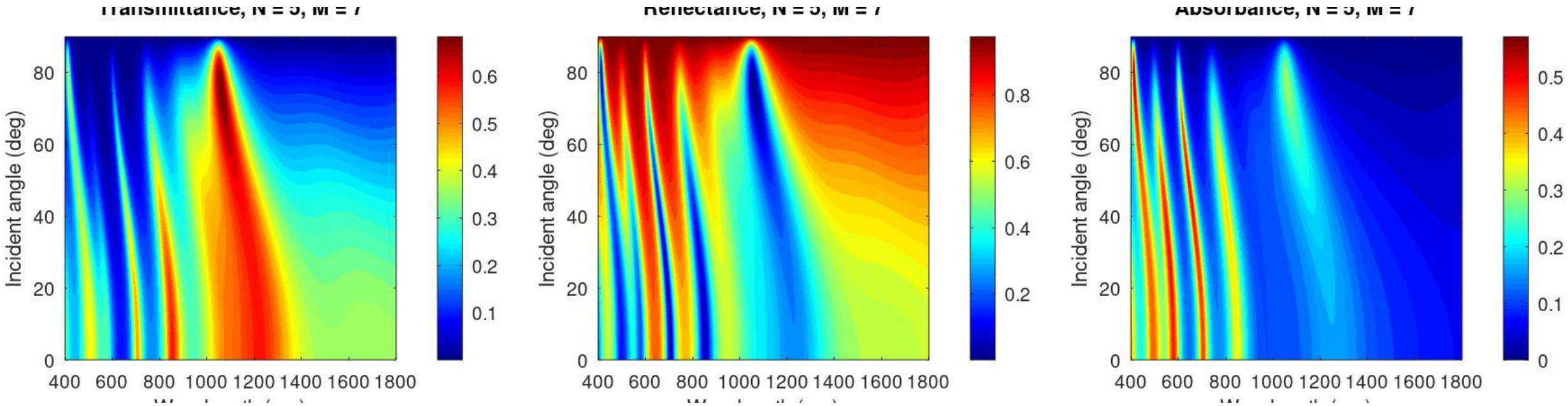
- Investigated wavelength: 400 – 1,800 nm
- Design of cavity geometry:
  - Sb_2Se_3 (80 nm): $n_{cavity} = 3.3 + 0.01i$ (amorphous)/$n_{cavity} = 4.1 + 0.25i$ (crystalline)
  - Si_3N_4 (100 nm): $n_{DBRH} = 2.0$
  - SiO_2 (100 nm): $n_{DBRL} = 1.45$
  - $N = 5$ / $M = 7$
- Superstrate is Air: $n_{air} = 1.0$
- Substrate is SiO_2: $n_{SiO2} = 1.45$
- Incident Angle: 0 – 90°



TRANSMITTANCE/REFLECTANCE/ABSORBANCE@AMORPHOUS



TRANSMITTANCE/REFLECTANCE/ABSORBANCE@CRYSTALLINE

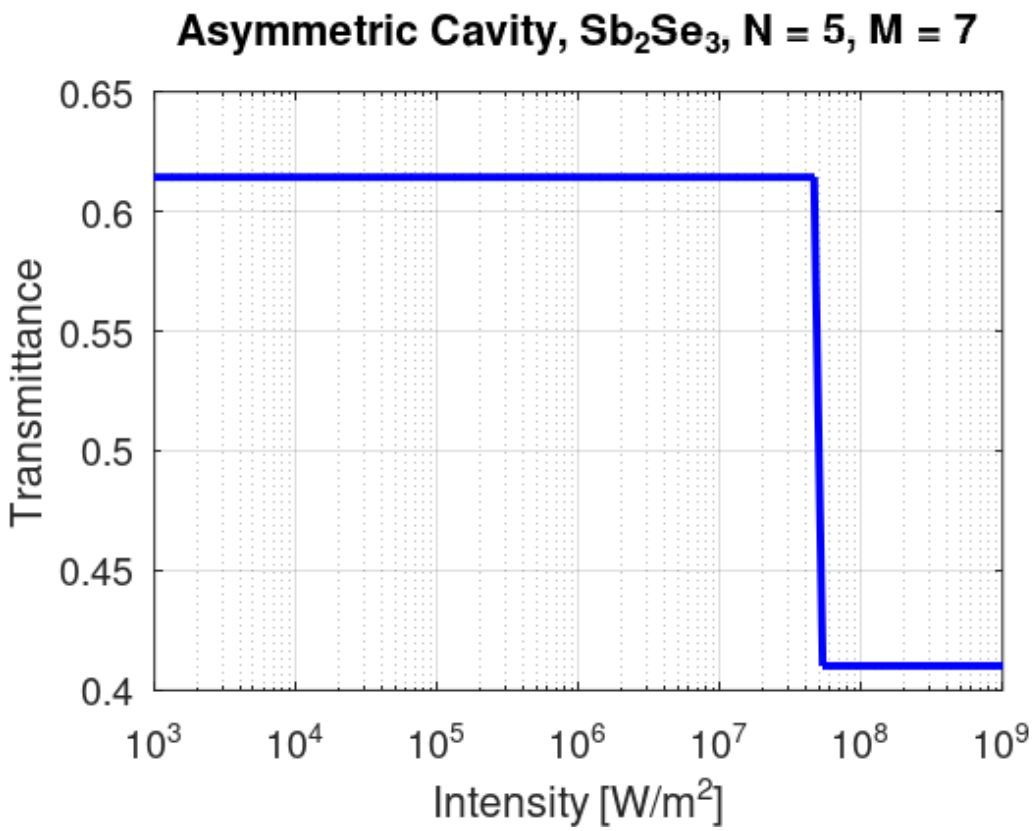


TRANSMITTANCE VS INTENSITY : OPTICAL LIMITING BEHAVIOR

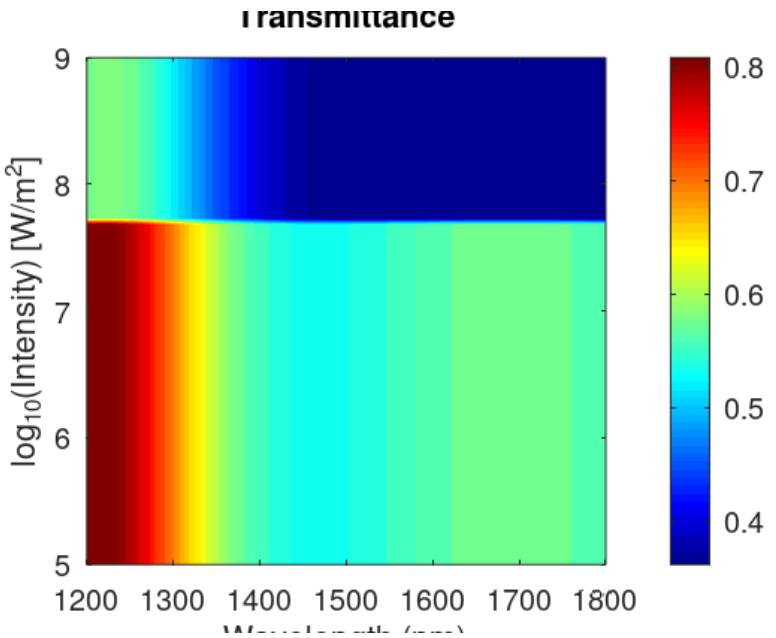
- Nonlinear Refractive Index Model:

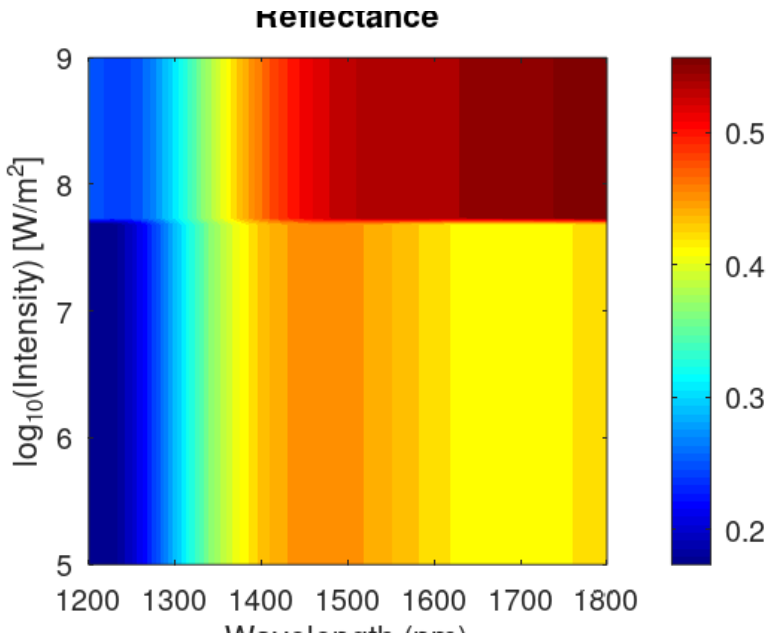
$$n_{PCM}(I) = \begin{cases} n_a + n_2 I_{in} & \text{for } I_{in} < I_{th} \\ n_c & \text{for } I_{in} > I_{th} \end{cases}$$

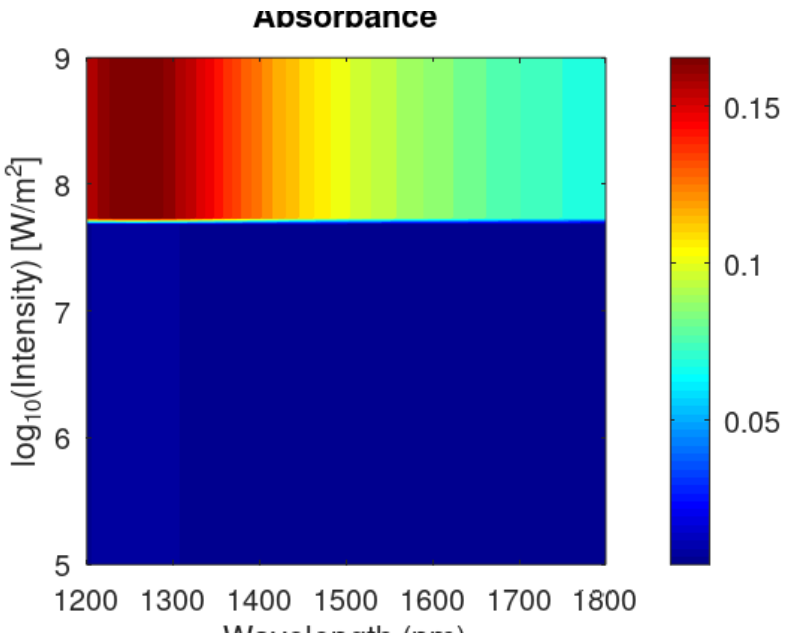
- n_2 is the Kerr-index (1×10^{-17} m^2/W)
- I_{th} is the threshold intensity (5×10^7 W/m^2)



TRANSMITTANCE/REFLECTANCE/ABSORBANCE VS INTENSITIES







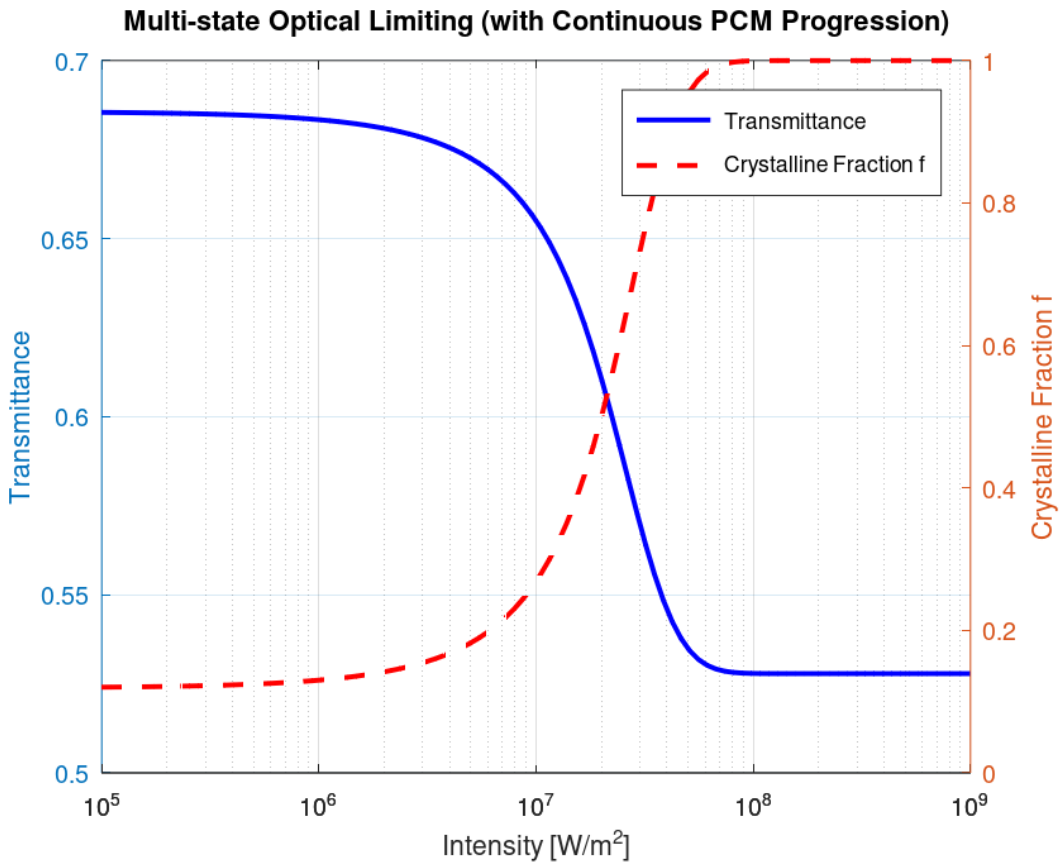
TRANSMITTANCE VS INTENSITY : OPTICAL LIMITING BEHAVIOR

- Multi-state model of PCM Refractive Index:

$$n_{PCM}(f) = n_a[1-f] + n_c f$$

$$f(I) = \frac{1}{1+e^{-k[I-I_{th}]}}$$

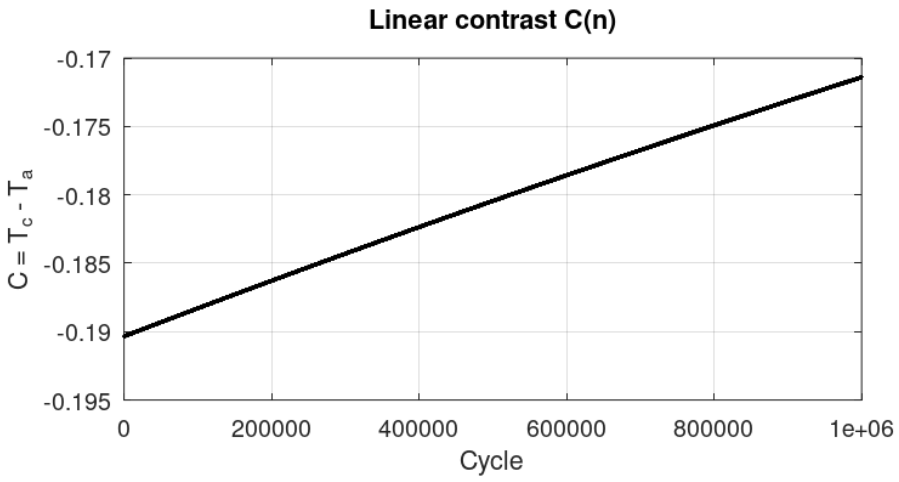
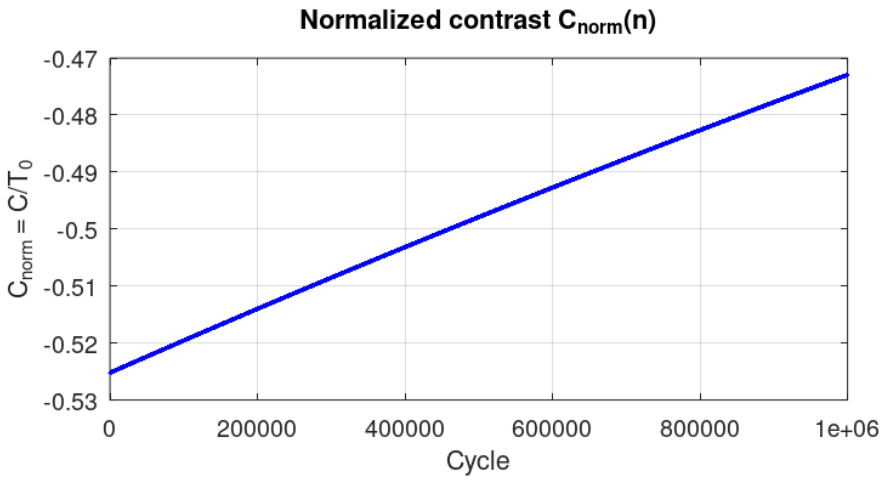
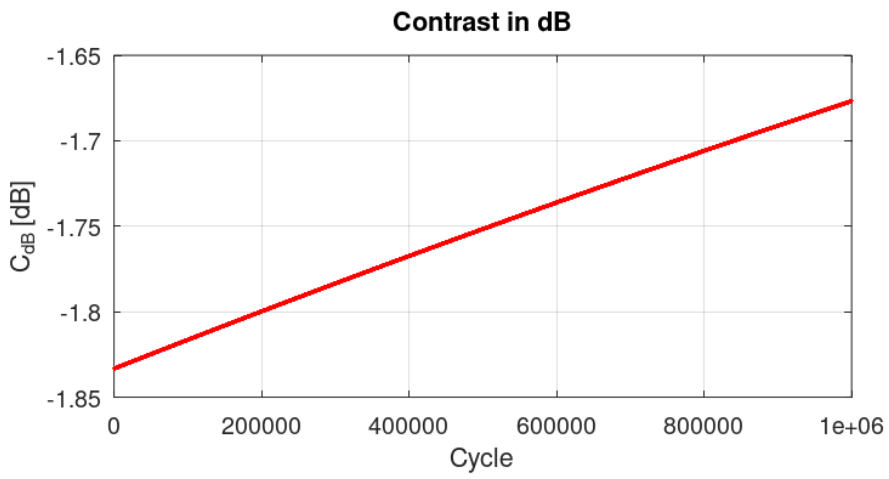
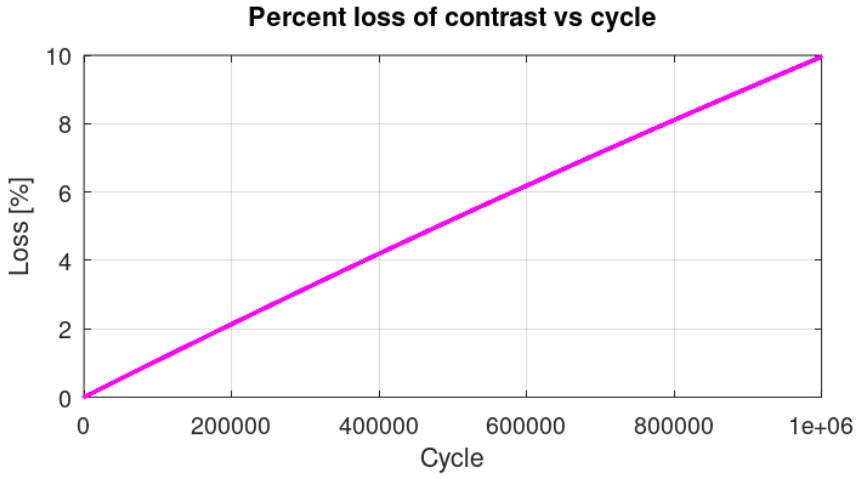
- f is the amorphous-to-crystalline fraction
- I is the optical intensity
- I_{th} is the threshold intensity (5×10^7 W/m^2)
- k is the transition slope ($k = 1$)



ENDURANCE DEGRADATION MODELS

- Exponential Decay Model: Refractive index of PCM reduces as

$$n_a(N) = n_a(0) + [n_c(0) - n_a(0)](1 - e^{-\alpha N})$$
$$n_c(N) = n_c(0) + [n_a(0) - n_c(0)](1 - e^{-\alpha N})$$
$$k_a(N) = k_a(0)[1 + \beta N]$$
$$k_c(N) = k_c(0)[1 + \beta N]$$

- N is cycle number (10^7 cycles)
- α is the degradation rate of real index (1×10^{-7} contrast reduction/cycle)
- β is the degradation rate of imaginary index (2×10^{-7} contrast reduction/cycle)

CONTRAST DEGRADATION @1,000,000 CYCLES